

# Exploring Satisfaction Level of Customers in Restaurants by using Latent Dirichlet Allocation(LDA) Algorithm

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**Abstract** – Social media has become an essential means for communicating the review/opinions of people around the world due to the rapid expansion and availability of the internet. Reviews and opinions are expressed both as text and audio. But text communication via networking media is overwhelming. Each and every second, a vast amount of information is produced online because of social media sites. However, online reviews on social media provides an excellent and trustworthy channel for examining the areas that require improvement and for understanding the needs of customers. This paper tries to understand the various topics that are discussed by customers about food with the aim of providing an insight to improve the area where there are negative comments. By doing this, customer retention will increase and gradually the business also. Topic modeling is done to find the hidden topic in a set of comments or customer reviews to find out on which topic customers are talking, thinking, or discussing more about restaurant services. It helps to read customers' mindsets, and accordingly, improvements can be made to increase consumer demand for a restaurant. So, a system is proposed to detect topics of textual reviews of different restaurants using the latent Dirichlet allocation (LDA) modelling approach. A probabilistic, statistical strategy for document designing called latent document analysis (LDA) identifies latent semantic topics in sizable corpora. The dataset used for this study was taken from Kaggle, which had 111,105 reviews. First, topics from reviews were extracted using the LDA model, then data was visualised using the LDAvis tool. The experiment's findings show that the analysis obtained a successful topic division outcome especially in long-to-short text level topic classification.

**Keywords** – Online restaurant reviews, topic modeling, Latent dirichlet allocation (LDA), stop words, bag of words

## I. INTRODUCTION

The contentment of a customer is the key to analysing if a service or a product satisfies customers' expectations. It is a crucial tool that offers an organisation significant insight into every aspect of their business, from the smallest to the largest. It helps a business reduce marketing costs, boost revenue, and attract more customers. Customer reviews or comments might help identify variables that weren't previously taken into

account, such as safe packaging, courteousness, the availability of customer service consultants, etc. It's crucial to give customers a sense of importance. By requesting their opinions about an item or service, this can be done. Customers feel valued and connected to a company when their thoughts and opinions about a good or service are sought after. Businesses have worked hard to learn what customers think of their goods and services. In many ways, social media has surpassed traditional communication in effectiveness. Via social media, users discuss about product or service, sharing their opinions and experiences. Topic modeling of customer reviews plays an important role as it finds the hidden topic inside the customer's opinion. It helps an organization to know what major topics customers use to discuss about the product or service of an organization. So, restaurants or organizations can focus on the fields that are the main topic of discussion among consumers.

Many research works were done on short texts, on song lyrics, on paragraphs to find the latent topics that exist in a corpus using various topic modeling algorithms such as LSA, BERT. But there is a limited amount of research work done for topic modeling on restaurant reviews, which are a mixture of long and short text reviews. Customer reviews of a company or restaurant can be a single word or a string of words. In a review, there are latent or hidden topics. A large number of review datasets fail to capture the individual nature of customers' feelings on different sides of an organization. So the topical discussion plays an important role. It finds the latent topics discussed by consumers and finds the majority of those topics, which helps organizations to know which things are being noticed by most of the consumers. And then improvement becomes easier. Techniques for topic modelling are particularly useful for revealing the hidden themes in customer evaluations.

A text document's semantic structure is determined by topic modelling, which may also arrange texts of various sizes. A topic model's topic probabilities give a clear indicator of the

document's understandability. This approach is applicable to domains like tendency analysis and document classification, among others. The topic modelling techniques for text classification are numerous.

## II. RELATED WORKS

LDA-based topic modelling system was proposed [1] and it has demonstrated great accuracy in the categorization of sentiment at the document-to-word level. Data visualisation is done with the LDAvis tool. To maintain the balance of the learning pace, improvement can be made by breaking score intervals up into multiple portions [1]. A system was proposed in which online product reviews of two competing products were used to analyze customer preferences. The key topic of customer reviews via text mining was done using LDA. It demonstrates the different topics of different products [2]. Using 3,302 internet evaluations of US-based eateries, topic modelling was applied and text score analysis was done on them. As a consequence, it revealed two new subjects about online ordering and home delivery, in addition to five previously undiscovered themes. Finding in-depth hidden topics can be done using a variety of topic modelling strategies [3]. To ascertain favorable, negative, and neutral reviews about restaurant policies, a real-time McDonald's dataset was analysed. The media portrayed the rude behavior of employees, so they wanted to comment on the rude service. A similar approach can be used to uncover the secret subject of customer conversation in different restaurant datasets [4]. A database of opinion-oriented words was used to predict aspects of specific sentiment on each review using latent Dirichlet allocation and compare the result to previous work in the same area. With the trade-off of slightly reduced accuracy, including seed words in the topic model can reduce the size of the feature set and result in quicker training duration for the classification algorithm [5]. In order to demonstrate the usage of topic modelling for researchers, a methodical strategy was offered to a case, and the LDA model was utilised in this instance. The following steps make up the framework: Subject modelling, pre-processing, and post-processing [6]. Two studies were proposed, one on Wikipedia articles and the other on topic modelling based on user tweets. Every step of the experiment procedure, including data collection, preprocessing of the data, and model training, is meticulously detailed and discussed. The paper's conclusion may be useful for social and commercial studies [7]. The research was conducted in an effort to identify emerging trends and paradigms that can be used as predictors for each issue. The journal's 3,269 research articles have been examined. With a 41% margin, the TF-IDF score-based model performed better [8]. The use of LDA for discrete data collecting was discussed in this paper. Topic probabilities offer a clear illustration of a file in text modelling. An analysis of collaborative filtering, text categorization, and document modelling has been conducted, comparing a concoction of the LSI and unigrams models was done [9]. LDAvis using LDA was presented in this paper, which was built by combining R and D3. A new technique

was proposed for choosing what word should be displayed to the user and also for presenting the result of a study. The visualization system LDAvis has been described, which helps users with flexible topic-term relationships [10]. A research was conducted on highly scholarly articles relating to topic modelling based on LDA between 2003 and 2016 in order to ascertain the progression of research, current trends, and intellectual topic modelling structure. To find ways to improve the models, many topic modelling techniques are discussed

[11]. Two categories were provided in this paper: the first one discussed the area of topic modeling methods, and the second one was a topic evolution model that considers an important time factor and also discusses different models [12]. Topic modeling related to 300 research articles was used for the survey in this paper. The paper includes different types of methods of topics modeling, evolution models of LDA etc and it has demands in various areas like software engineering, scientific writing. For a deeper grasp of topic modeling, a quantitative assessment of topic modeling methodologies is presented in [13]. An article was presented which discussed briefly the process and the logic of topic modeling(TM). Transcripts of U.S. presidential discussions between Donald Trump and Hillary Clinton of 2016 have been analyzed using latent semantic analysis technique. The result showed the effectiveness of TM in locating thematic patterns hidden inside an enormous amount of text [14]. An article on short text topic modeling(STTM) was presented which described a survey and classification of STTM algorithms for topic modeling on short term texts. The article also describes the advantages and disadvantages of STTM methods. The performance comparison of the quality of the topic of different STTM techniques was done using 2 real world datasets. At the end, the facing challenges and future work scope in this field was discussed deeply [15]. A research as done to find the topic from essays written by school kids used latent dirichlet allocation model to find the understanding of students about the surroundings. This text analysis confirms the manual testing of teachers. As a potential direction for future work, socio-scientific problem-solving exercises can be created to assess conceptual validity and create dynamics that can inspire and empower students to think critically about science [16].

## III. METHODOLOGY

In order to carry out the analysis a dataset containing reviews about restaurants was collected from Kaggle [19]. Preprocessing was done on the dataset before training. The workflow is shown in Fig. 1, and the others are detailed below.

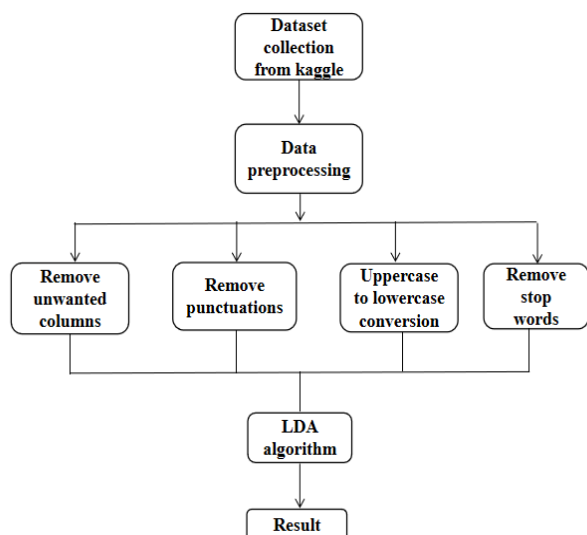


Fig-1 Work flow diagram

For topic modeling, before using LDA, usually data pre-processing is performed to clean and transform the text into a suitable format for the algorithm. Steps of data pre-processing before applying LDA are - removing symbols and punctuation, tokenization, remove stop words, data stemming, vectorization.

#### A. Preparation of dataset

The process of convolving, estimating, and checking the precise insights for research based on a given system is known as data gathering. Data was gathered from Kaggle. The total number of reviews was 111,105. The data was in the form of long and short paragraphs or just a word. After collecting the data, unwanted columns, "SNo, name, rating" have been deleted, punctuation has been removed, and conversion of uppercase to lowercase characters has been done. Preprocessing of the text is carried out prior to use of the algorithms. This transforms the raw information into a more manageable and efficient format. Data cleaning, which deals with punctuation, stop words, repetitive characters, lower case speech, etc., is one of many pre-processing procedures. Four columns make up the dataset: SNo, name, rating, and review.

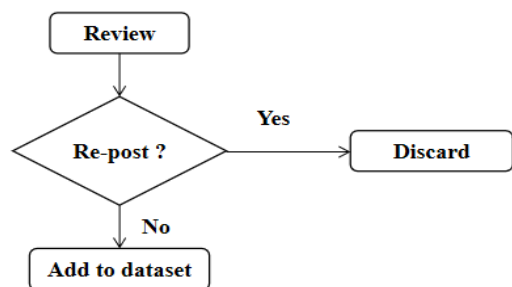


Fig-2 Selection process of reviews

#### A.1 Delete all punctuation

With the input provided, interesting keywords are in demand so that processing may be done on them. Punctuation marks in the input structure must be deleted because they are insignificant tokens. One strategy would be to partition the input into words based on white space using string translation, then replace all punctuation with periods. Punctuation will be taken out of tokens after they have been received. In order to do this, the list of punctuation was obtained using the Python string module, and punctuation was then removed using a token-by-token loop.

#### B.2 Character repetitions

People on social media no longer strictly adhere to grammar rules. They express their feelings by employing alternative word spellings and abbreviations, such as woww, coooool, etc. Users used to type things like "It was toooo good" in order to emphasize the word "too". Computers, however, do not comprehend that "toooo" is one of the variants of "too". Thus, it is necessary to convert such forms into meaningful words. By removing these bothersome repeating characters, a suitable, meaningful word that can be found in the English dictionary is produced.

#### C.3 Stop words

Stop words are those undesirable terms that only take up extra space in the database and slow down processing. Several systems use different stop terms. The words "some," "can," "these," "against," "at," "because," "is," "just," etc are among the stop words used in English.

To remove stop words from text, first NLTK library was imported and stop words were downloaded. Then stop word corpus was imported and stop words were defined for the text. The text input was next tokenized using the NLTK word-tokenize function, and stop words were then eliminated from the tokenized text using list inspection.

#### B. Model

There are several techniques, such as Latent Semantic Analysis(LSA), Parallel Latent Dirichlet Allocation (PLDA), Non Negative Matrix Factorization(NMF) etc for topic modeling. A lot of data can be compressed with LDA. One can use this topic modelling technique to find the key points being made in the documents rather than going through every detail in each one. This makes it simple to understand the text data's underlying themes. Additionally, LDA may detect subjects without labelled data. Unsupervised learning algorithms can be used to a variety of text data without the need for manual annotation. Moreover, LDA may be customised to incorporate prior information and be utilised with a variety of text representations, such as bags of words. In order to perform topic modeling on 111,105 reviews, LDA algorithm was used. The step by step process is shown below in the diagram in Fig- 3.

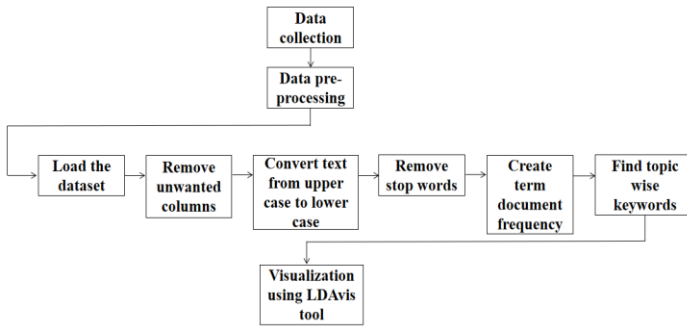


Fig- 3 Process to the visualization

### C. LDA

LDA is a well-liked topic modelling method for isolating themes from a section given. Latent refers to anything which already exists but isn't fully developed. The latent translates as concealed or hidden. The themes that we wish to get data from are secret topics. It hasn't been found yet. Following the distribution and processing of Dirichlet, comes the Dirichlet allocation.

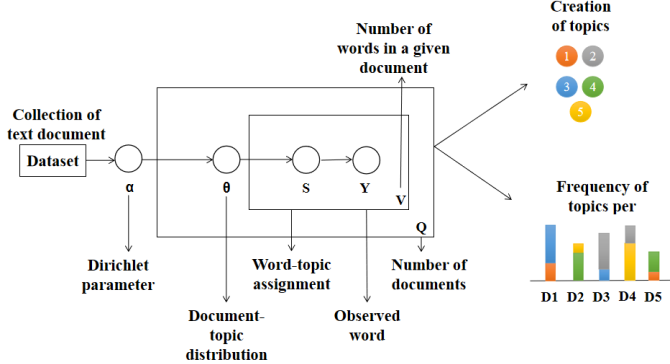


Fig-4 LDA model [18]

Groups of words with statistical significance within a passage are referred to as topics or themes. Without getting into the specifics of the procedure, the dirichlet model defines the word pattern that are commonly appearing, repeated, and comparable to one another in the context. When it comes to topic modelling, the method helps determine how likely it is that words strewn across the content will appear again. This gives the algorithm the ability to produce data values and compute probabilities. The LDA bases two critical hypotheses on- papers cover a variety of subjects, and the subjects range from works to tokens.

Cleaning, preprocessing, and tokenizing the text data are the first steps in the process. The following document word matrix will be obtained after preprocessing the documents, and it will include:

- D1, D2, D3, D4 and D5 are 5 documents
  - The words represented by the  $Y_n$ , says there are 10 special phrases from  $Y_1$  to  $Y_{10}$ .
- So, the matrix's design is  $5 \times 10$ , which is 5 rows and 10 columns.

|    | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 | Y8 | Y9 | Y10 |
|----|----|----|----|----|----|----|----|----|----|-----|
| D1 | 0  | 1  | 1  | 1  | 0  | 1  | 0  | 1  | 0  | 1   |
| D2 | 1  | 1  | 1  | 0  | 1  | 1  | 1  | 1  | 1  | 1   |
| D3 | 1  | 0  | 0  | 0  | 1  | 0  | 1  | 1  | 1  | 0   |
| D4 | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0   |
| D5 | 1  | 1  | 1  | 1  | 0  | 1  | 0  | 0  | 1  | 1   |

Fig-5 Document Word Matrix[17]

Now, mainly in the lexicon is above the pre-processed word document pattern; every row on the table is a document, and every column represents words or tokens. Using LDA, The term matrix for the topic and the document are both listed below. The document-word structure is divided into two matrices.

|    | L1 | L2 | L3 | L4 | L5 | L6 | L7 | L8 |
|----|----|----|----|----|----|----|----|----|
| D1 | 1  | 0  | 1  | 0  | 0  | 1  | 0  | 1  |
| D2 | 0  | 1  | 0  | 0  | 1  | 0  | 1  | 1  |
| D3 | 1  | 1  | 0  | 1  | 0  | 0  | 0  | 0  |
| D4 | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 0  |
| D5 | 1  | 0  | 1  | 0  | 1  | 1  | 1  | 1  |

Shape  $5 \times 8$

|    | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 | Y8 | Y9 | Y10 |
|----|----|----|----|----|----|----|----|----|----|-----|
| D1 | 0  | 1  | 1  | 1  | 0  | 1  | 0  | 1  | 0  | 1   |
| D2 | 1  | 1  | 1  | 0  | 1  | 1  | 1  | 1  | 1  | 1   |
| D3 | 1  | 0  | 0  | 0  | 1  | 0  | 1  | 1  | 1  | 0   |
| D4 | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0   |
| D5 | 1  | 1  | 1  | 1  | 0  | 1  | 0  | 0  | 1  | 1   |

Shape  $5 \times 10$

|    | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 | Y8 | Y9 | Y10 |
|----|----|----|----|----|----|----|----|----|----|-----|
| L1 | 0  | 1  | 1  | 1  | 0  | 1  | 0  | 1  | 0  | 1   |
| L2 | 1  | 1  | 1  | 0  | 1  | 1  | 1  | 1  | 1  | 1   |
| L3 | 1  | 0  | 0  | 0  | 1  | 0  | 1  | 1  | 1  | 0   |
| L4 | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0   |
| L5 | 1  | 1  | 1  | 1  | 0  | 1  | 0  | 0  | 1  | 1   |
| L6 | 0  | 1  | 0  | 1  | 0  | 1  | 1  | 0  | 1  | 0   |
| L7 | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 0  | 1  | 1   |
| L8 | 1  | 0  | 0  | 1  | 1  | 1  | 0  | 1  | 0  | 1   |

Shape  $8 \times 10$

Fig-6 Conversion of Document Word Matrix[17]

The subjects that could be in the document are represented by L in the document-topic matrix. Here, a total of 5 documents and 8 subjects are covered. The matrix has an  $8 \times 5$  dimension.

The words that those themes may contain are in the word-topic group. Here, 8 topics and 10 distinct values are present, so, the matrix had the shape of  $8 \times 10$ .

The graphic below shows how the complete LDA space and associated dataset are represented-

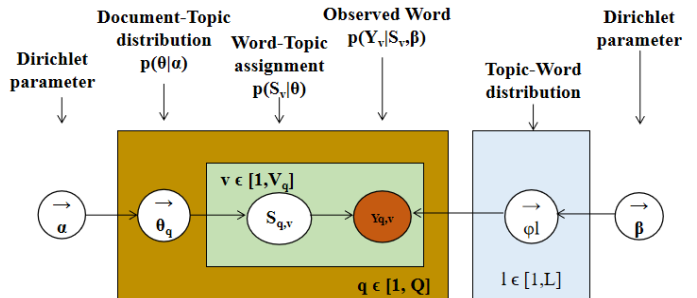


Fig-7 Entire space of LDA and dataset representation[17]

- In the diagram, the Brown box presents all documents represented by Q. In this case there are 5 documents,  $Q=5$ .

- II. The green box in the diagram shows how many words there are presented in a particular document given by V.
- III. There may be any number of words inside the green box. Y is one of those words, and it's visible in the orange circle.

Each phrase has a connection to a secret topic that is represented as S, claims LDA.  $\Theta$ , which in this instance reflects the relationship of Z to a topic word in these records, is used to calculate the spread of topical terms in the paragraph.

The distribution is controlled by two factors in the LDA framework:

- a) Topic distribution is controlled by Alpha and
- b) Beta controls for subject-specific word distribution.

#### Summarize

- The collection has Q documents in total.
- The total word count in a document is V.
- The file contains the word Y.
- S is the word's designated secret topic.
- The topic distribution is called  $\Theta$ .
- Alpha( $\alpha$ ) and Beta( $\beta$ ) are LDA model dimensions.

#### C.1 Optimization of LDA distribution

In order to determine distribute document topics and topic words in the most efficient way possible, LDA seeks to identify the most ideal document topic matrix and topic word matrix representations. LDA implies that topics and terms within subjects are mixed in documents. In order to determine which subjects and which words would have formed the specific document, LDA goes back to the document level. There are 5 documents in this corpus, numbered D1 through D5, each with a specific number of words [17]-

- $D1 = (Y1, Y2, Y3, Y4, Y5, Y6, Y7, Y8, Y9, Y10)$
- $D2 = (Y'1, Y'2, Y'3, Y'4, Y'5, Y'6, Y'7, Y'8, Y'9, Y'10, Y'11, Y'12)$
- $D3 = (Y''1, Y''2, Y''3, Y''4, Y''5, Y''6, Y''7, Y''8, Y''9, Y''10, Y''11, Y''12, Y''13, Y''14, Y''15, Y''16, Y''17)$
- $D4 = ((Y'''1, Y'''2, Y'''3, Y'''4, Y'''5, Y'''6, Y'''7, Y'''8, Y'''9, Y'''10, Y'''11, Y'''12, Y'''13, Y'''14, Y'''15))$
- $D5 = (Y''''1, Y''''2, Y''''3, Y''''4, Y''''5, Y''''6, Y''''7, Y''''8, Y''''9, Y''''10, Y''''11, Y''''12, Y''''13, Y''''14, \dots, Y''''40, Y''''41, Y''''42)$

#### C.2 LDA's earliest iteration-

The topic is initially assigned at random to each and every word in the text. The letter L stands for topics. The words in the corpus will be linked to a variety of themes at random, as displayed below [17]-

- $D1 = (Y1 (L4), Y2 (L2), Y3 (L5), Y4 (L7), Y5 (L1), Y6 (L8), Y7 (L3), Y8 (L6), Y9 (L1), Y10 (L5))$

- $D2 = (Y'1 (L2), Y'2 (L1), Y'3 (L4), Y'4 (L6), Y'5 (L7), Y'6 (L8), Y'7 (L3), Y'8 (L5), Y'9 (L2), Y'10 (L5), Y'11 (L7), Y'12 (L8))$
- $D3 = (Y''1(L4), Y''2 (L8), Y''3 (L2), Y''4 (L6), Y''5 (L1), Y''6(L3), \dots, Y''15 (L8), Y''16(L2), Y''17 (L5))$
- $D4 = (Y'''1(L6), Y'''2 (L4), Y'''3 (L5), Y'''4 (L3), Y'''5 (L1), Y'''6 (L8) \dots, Y'''12 (L2), Y'''13 (L8), Y'''14 (L7))$
- $D5 = (Y''''1 (L3), Y''''2 (L6), Y''''3 (L1), Y''''4 (L5), Y''''5 (L1), Y''''6 (L8), Y'7 (L2) \dots, Y''''30(L5), Y''''32(L2), Y''''33 (L6))$

The result is output in the form of documents that are composed of subjects and topics that are composed of words. Documents cover a variety of subjects [17]-

- $D1 = L4 + L2 + L5 + L7 + L1 + L8 + L3 + L6 + L1 + L5$
- $D2 = L2 + L1 + L4 + L6 + L7 + L8 + L3 + L5 + L2 + L5 + L7 + L8$
- $D3 = L4 + L8 + L2 + L6 + L1 + L3 + \dots + L8 + L2 + L5$

The topics are a mixture of words [17]

- $L1 = Y2 + Y5 + Y'7 + Y'8 + Y'12 + Y''5 + Y''4 + \dots + Y''11 + Y''15 + Y'''3 + Y'''8$
- $L2 = Y2 + Y'1 + Y'6 + Y''12 + \dots + Y'''5 + \dots$
- $L3 = Y5 + Y'7 + Y''1 + Y''2 + Y''12 + Y'''7 + Y'''3 + \dots + Y'''12 + Y'''30 + \dots$

LDA will also provide word combinations for other other themes.

#### C.3 After LDA's initial implementation

LDA offers the basic text topic and topic-word matrices following the initial iteration. These findings must be optimized, which LDA does by going over all of the documents and terms repeatedly. LDA assumes that all of the allocated topics—aside from the current word—are accurate. LDA cycles through each word 'Y' and each document 'D'. For each topic (L), it calculates two probabilities, p1 and p2, where -

- P1 shows the portion of the document (D) that are currently devoted to the content (L)
- P2 is the proportion of tasks in all texts obtained from the term Y that are linked to the topic (L).

The equations for p1 and p2 are-

- $P1 = \text{proportion}(\text{topic } L / \text{document } D)$
- $P2 = \text{proportion}(\text{word } Y / \text{topic } L)$

By combining the probabilities p1 and p2, LDA generates a new probability. The two probabilities were combined to create this new probability. Based on this product likelihood, LDA selects a new topic that is the most pertinent topic for the present word. In order to select a new topic 'L' LDA runs numerous iterations until it reaches a steady state. A term-document combination and a term-word combination are most effectively represented at the LDA convergence point. This is how LDA conducts its business.



#### IV. EXPERIMENTAL RESULTS

##### A. Dataset description

A dataset of 111,105 reviews has been used here. Datasets have been collected from Kaggle. A lot of duplicate reviews were present there. Data pre-processing is done to remove duplicates. The total number of unique reviews is 111,105. The dataset contains 4 columns: named- SNo, name, rating, and review.

| SNo | name     | rating | review                                            |
|-----|----------|--------|---------------------------------------------------|
| 1   | 1000 B.C | 1.0    | One of the best place in Indira Nagar undoubte... |
| 2   | 1000 B.C | 4.0    | Loved the place, the ambience is great for an ... |
| 3   | 1000 B.C | 1.0    | Everything about this place is great service w... |
| 4   | 1000 B.C | 5.0    | Nice place decent crowd and courteous staff. B... |
| 5   | 1000 B.C | 1.0    | Tried this place for the first time had heard ... |

Fig-8 Dataset

##### B. Remove extra columns

Next, the unwanted columns of the dataset, named SNo, name, and rating, have been removed because the column "review" will be the only one used for the work. To do so, pandas library has been used. Then an excel dataset is loaded into Pandas DataFrame. The identification of columns which had to be removed was done. The columns are then removed using the drop method of the DataFrame. Using the to\_excel method, the modified dataset was saved. It is shown in Fig-9.

| review                                            |
|---------------------------------------------------|
| One of the best place in Indira Nagar undoubte... |
| Loved the place, the ambience is great for an ... |
| Everything about this place is great service w... |
| Nice place decent crowd and courteous staff. B... |
| Tried this place for the first time had heard ... |

Fig- 9 Column "review"

##### C. Punctuation deletion

Processing can be done only on keywords from the given input. Punctuations are uninteresting tokens that have to be removed before processing data. Next, the punctuation from reviews has been removed. To remove punctuation from the Pandas library string module was used. A function was defined and applied to the cells in the dataset using a method called applymap. The modified dataset is then saved as shown below in Fig-10.

one of the best place in indira nagar undoubtedly service is top notch and i would suggest they should levy service charge chicken lol  
scented chicken was awesome amritsari fish was ok and can be improved tandoori murgh tikka pizza was very delicious witty wit bee  
good with passion fruit flavor overall a very good place to visit again  
loved the place the ambience is great for an afternoon lunch as well as an evening get together the food was delicious and the service  
taking care of our table and we loved his attentiveness and service would definitely be visiting this place again and recommend this to  
everything about this place is great service was excellent very tasty food its huge so easy to find a table just a little high on price my re  
nice place decent crowd and courteous staff but the food was not upto my expectations nachos lacked cream and cheese paneer tik  
jalapeno cheese balls and the drinks were good  
tried this place for the first time had heard about the appetizers and the kebabs the starters are subdivided into asian international and  
international however was really good as compared to indian bbq chicken sausage pizza was delicious must try

Fig- 10 Reviews with no punctuation

##### D. Stop word deletion

Stop words are undesirable extra-large words that delay down database operations. To remove stop words from text, first NLTK library was imported. Then stop words were downloaded. The stop word corpus was then imported and stop words were defined for the text. The NLTK word\_tokenize function was used to tokenize the input text after that, and list inspection was used to remove stop words from the tokenized text. Reviews, which are displayed in Fig-11 below, no longer contain any stop words.

'one', 'best', 'place', 'indira', 'nagar', 'undoubtedly', 'service', 'top', 'notch',  
'would', 'suggest', 'levy', 'service', 'charge', 'chicken', 'lollipop', 'good',  
'signature', 'dish', 'jalapeno', 'scented', 'chicken', 'awesome', 'amritsari', 'fish',  
'ok', 'improved', 'tandoori', 'murgh', 'tikka']

Fig- 11 Reviews without stop words

##### E. Term document frequency creation

Then the dictionary was created. From the dictionary, the creation of a corpus and then the term document frequency have been created. The feature turns a dictionary into a collection of words. To do so sklearn library was imported and from from sklearn library, TfidfVectorizer was imported. The dataset was then loaded into Pandas DataFrame. A list was created of all text documents and an TfidfVectorizer object was used in order to tokenize the text document. The TF-IDF matrix then saved. To create the TF-IDF matrix, the fit\_transform method was used. As illustrated in Fig-12, the outcome is a list of matrices, each containing a collection of tuples, with a length equal to the number of documents.

[(0, 1), (1, 2), (2, 1), (3, 1), (4, 1), (5, 2), (6, 1), (7, 1), (8, 1), (9, 1), (10, 1),  
(11, 3), (12, 1), (13, 1), (14, 1), (15, 1), (16, 1), (17, 1), (18, 1), (19, 1), (20, 1),  
(21, 1), (22, 1), (23, 1), (24, 1), (25, 1), (26, 1), (27, 2), (28, 1), (29, 2)]

Fig-12 Term document frequency

##### F. Find keywords for each topic

Then the process of finding keywords for topics is done. Gensim library was used for this process. A dictionary was created of all the unique words in the dataset and conversation was made into a bag of words. Then training the LDA model was done by using the corpus. Then the print\_topic method was used to print top keywords of topics. A comma separates each created topic from the next. In Fig-13, the three terms from each topic that are most likely to appear are shown.

```
[('0.016*"ordered" + 0.010*"pav" + 0.009*"sandwich" + 0.008*"sweet" + '
'0.007*"order" + 0.006*"vada" + 0.006*"fresh" + 0.006*"juice" + 0.005*"time"
'+ 0.005*"like"'),
(1,
'0.056*"chicken" + 0.027*"good" + 0.020*"biriyani" + 0.019*"ordered" + '
'0.018*"rice" + 0.012*"food" + 0.012*"paneer" + 0.012*"taste" + '
'0.010*"biriyani" + 0.009*"veg"'),
(2,
'0.059*"food" + 0.042*"good" + 0.027*"taste" + 0.024*"indian" + '
'0.016*"place" + 0.015*"quality" + 0.013*"quantity" + 0.012*"try" + '
'0.012*"also" + 0.012*"price"'),
```

Fig-13 Keywords of the topics

### G. Result

A performance comparison of different types of configurations for processing long and short text data has been done. It starts with understanding the reason for a review of the topic model, which is crucial. Existing techniques have been reviewed and only barely touched the subject of coherence, besides the readily available coherence measurements. The LDA model has been built using Gensim. The LDAvis tool has been used for visualization purposes. A total of ten topics have been created, with each topic displaying the top 30 most relevant terms (23.8% of tokens). The result is shown below in Fig-14.

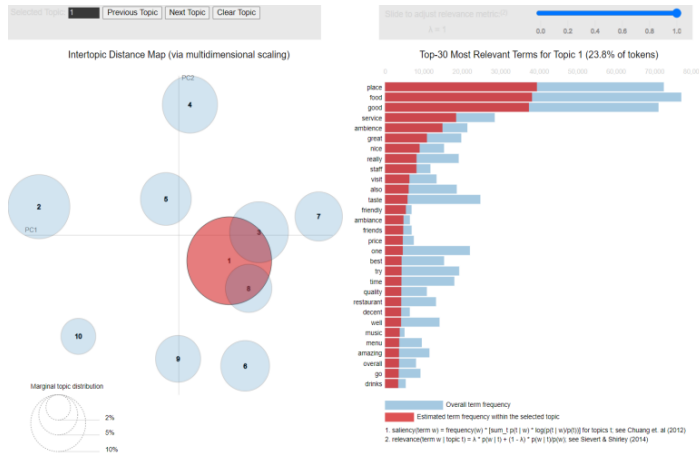


Fig-14 Ten Topics and top thirty terms of topics

## V. CONCLUSION

In this paper, a LDA model has been presented for topic modeling of customers' reviews of restaurants. The LDA model gives accurate results for long as well as short texts. It benefits from the ability of LDA to perform feature extraction and find the most relevant terms for various topics.

The model was trained on a dataset of 111,105 reviews collected from Kaggle. Data pre-processing has been done before applying the data as input to the algorithm. As the model separates topics and finds the most discussed terms of the topics, the accuracy of the result increases. The key to boosting success is to cut down on the number of dictionary words. Words with a weight below the threshold value are eliminated from the dictionary throughout this phase. So that as the stage increases, the importance of connected terms rises.

First, theme elicitation has been done. The number of concealed topics is calculated by maximum information divergence over all themes. In the topic model, the conditional probability of words has been given a particular topic.

The 10 extracted themes exhibit topics that are distinctly different from one another; there are relationships between each set of topics. The connections between the different subjects make it easier for researchers and business professionals to understand and quantify the various aspects of the customer experience.



Fig-15 Wordcloud of 10 topics

In future work, the proposed model could be adapted for Twitter data, Instagram data, or data from Facebook on different restaurants to find out the most discussed topic by customers about the restaurant, and topic modeling can be applied for classification of various types of positive data and also in various types of negative data. It will help the organization's members understand where they have to focus more in order to provide a high-quality service.

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